

Boyuan Chen

San Francisco, CA * Home Page: boyuan.space * boyuanc@mit.edu

EDUCATION

Massachusetts Institute of Technology (MIT), PhD student in EECS, MIT CSAIL

2021 - 2025

UC Berkeley, BA Computer Science (EECS Honor Class), Applied Math, with distinction

2017-2021

My research centers on generative world model and associated decision-making algorithms. Towards this goal, I published a focused set of works bridging autoregressive diffusion, RL via inference, and robotics. My works are widely adopted by industry as the standard paradigm for autoregressive video models as well as autonomous agents on these world models.

EXPERIENCE

OpenAI

Member of Technical Staff (Research)

Jun 2025 - present

- Research self-improving AI in the Recursive Self Improve (RSI) team.
- Former member and one of the earliest researchers of the GPT Image team.
- Key contributor of GPT Image 1.5, GPT Image 2 and GPT Image 2.5, OpenAI's state-of-the-art image generation model.
- Featured in OpenAI's GPT image 2 livestream with Sam Altman, presenting advanced text-rendering capability.
- Contributor of GPT 5, GPT 6 series and OpenAI robotics.

Google DeepMind

Research Intern

May 2023 - Aug 2023

- Lead the training of a multi-modal Large Language Model (MLLM) with large scale synthetic data.
- Implemented the entire data synthesis pipeline, instruction tuning that was later adopted into Gemini 2.0.

Google X (or X, the Moonshot Factory)

AI resident

May 2022 - Aug 2022

- Develop machine learning algorithms for sequential decision making in robotics
- Visual grounding for Large Language Model

MIT Computer Science and Artificial Intelligence Laboratory (CSAIL)

Researcher

Sep 2021 - present

- Machine learning research advised by Prof. Vincent Sitzmann and Prof. Russ Tedrake
- Research focus: Generative world model, Model-based RL, Foundation models for decision making, Robotics.

Berkeley Artificial Intelligence Research Lab

Researcher

Jan 2019 - Aug 2021

- Reinforcement learning research with Prof. Pieter Abbeel; Computer vision research Prof. Trevor Darrell
- Student researcher on unsupervised learning, 3d vision, visual reinforcement learning and generalizable manipulation.

SKILLSET

Language & Tools: Python, C++/C, Java, PyTorch, Tensorflow, Jax, Pax, Flume, OpenCV, MongoDB, TensorRT, ZeroMQ, Qt5

Machine Learning: Deep Reinforcement Learning, Generative Models (Diffusion, Flow, GAN, VAE), Variational Inference, Time Series Prediction, Planning Imitation Learning, World Model, Large Language Model, Multimodal Model, Data Synthesis.

PUBLICATION

Large Video Planner Enables Generalizable Robot Control

Arxiv 2025

B. Chen*, T. Zhang*, H. Geng*, K. Song, C. Zhang, P. Li, W. T. Freeman, J. Malik, P. Abbeel, R. Tedrake, V. Sitzmann, Y. Du. (*Equal Contribution)

Diffusion Forcing: Next-token Prediction Meets Full-Sequence Diffusion

NeurIPS 2024

B. Chen, D. Monso, Y. Du, M. Simchowitz, R. Tedrake, V. Sitzmann.

Spatial VLM: Endowing Vision-Language Models with Spatial Reasoning Capabilities

CVPR 2024

B. Chen, Z. Xu, S. Kirmani, B. Ichter, D. Driess, P. Florence, D. Sadigh, L. Guibas, F. Xia

Self-Supervised Reinforcement Learning that Transfers using Random Features

NeurIPS 2023

B. Chen, C. Zhu, P. Agrawal, K. Zhang, A. Gupta

History Guided Video Diffusion

ICML 2025

K. Song*, B. Chen*, M. Simchowitz, Y. Du, R. Tedrake, V. Sitzmann. (*Equal Contribution)

Open-vocabulary Queryable Scene Representations for Real World Planning

ICRA 2023

B. Chen, F. Xia, B. Ichter, K. Rao, K. Gopalakrishnan, M. Ryoo, A. Stone, D. Kappler

Unsupervised 3d Keypoint Learning for control B. Chen, D. Pathak, P. Abbeel.	ICML 2021
Discovering Diverse Multi-Agent Strategic Behavior via Reward Randomization Z. Tang, C. Yu, B. Chen, H. Xu, X Wang, F. Fang, S. Du, Y. Wang, Y. Wu	ICLR 2021
DittoGym: Learning to Control Soft Shape-Shifting Robots S. Huang, B. Chen, H. Xu, V. Sitzmann	ICLR 2024
Reasoning or Reciting? Exploring the Capabilities and Limitations of LLM Through Counterfactual Task Z. Wu, L. Qiu, A. Ross, E. Akyürek, B. Chen, B. Wang, N. Kim, J. Andreas, Y. Kim	NAACL 2024
Extraneousness-Aware Imitation Learning R. Zheng, K. Hu, B. Chen, H. Xu.	ICRA 2023
Zero-shot Policy Learning with Spatial Temporal Reward Decomposition on Contingency-aware Observation B. Chen*, H. Xu*, Y. Gao, T. Darrell.	ICRA 2020
Unifying 3D Representation and Control of Diverse Robots with a Single Camera S. Lester Li, A. Zhang, B. Chen, H. Matusik, C. Liu, D. Rus, V. Sitzmann	Nature

ACADEMIC SERVICE

Reviewer of NeurIPS, ICLR, ICML, CVPR, RSS, ICRA, IROS, RAL, AAAI	<i>2021-present</i>
Teaching Assistant, MIT 6.S183 Diffusion Models, 6.4210/6.4212 Robotic Manipulation	<i>Sep 2022 – Dec 2022</i>

PERSONAL PROJECTS

Generative Boba, a diffusion-themed boba shop at MIT CSAIL	<i>Mar 2024 – May 2025</i>
Autonomous multi-floor food delivery robot (Control, Planning, Sensing, Vision, ROS)	<i>Sep 2019 - Dec 2019</i>
ICRA Robomaster AI Challenge Autonomous Combat Robot (Vision, Planning, ROS, Control, AI)	<i>Jan 2019 - May 2019</i>
Personal drivable RC robot (CAD, Manufacture, Electronics, Control, Embedded System)	<i>May 2019 - Aug 2019</i>
Autonomous Multi-Terrain Rover (CAD, Manufacture, Electronics, Computer Vision, Planning)	<i>Oct 2017 - Aug 2018</i>
Autonomous Tracking Drone (Computer Vision, Embedded System)	<i>Sep 2016 - Aug 2017</i>

HONOR

Seneff-Zue CS Fellowship	<i>Feb 2021</i>
Winner, Facebook Pytorch Summer Hackathon	<i>Aug 2019</i>
Finalist, ICRA 2019 Robomaster AI Challenge	<i>May 2019</i>
Winner, Record Keeper, UC Berkeley CS 61C Neural Network Inference Optimization Contest	<i>Aug 2018</i>
Winner, CS170 Efficient Algorithms Contest	<i>Oct 2018</i>
2 nd place, Google Puzzlehunt, second fastest prize eligible team out of 800+ teams of Google employees	<i>Aug 2022</i>
Honor degree in EECS, High honor in general scholarship, Dean's List, UC Berkeley	<i>2017-2021</i>